Air Pollution in Germany

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Agenda

- Introduction
- Data Collection
- Challenges
- Research Questions
- Website

Introduction

Introduction

- Air Quality in Germany
- How does the quality change based on location/time/pollutant?
- Pollutants:
  - PM_{10} - Fine particles (≤10 μm) from traffic, industry, and dust that can enter the lungs and impact health.
  - $PM_{2.5}$ -Very fine particles (≤2.5 μm) from combustion (e.g., vehicles, heating, power plants) that can reach deep lungs and enter the bloodstream.
  - NO_2 - A gaseous pollutant from combustion that irritates the respiratory system and contributes to ozone and particulate formation.

Research Questions - UBA

- How did the average values of PM_{10}, $PM_{2.5}$ and NO_2 change in Germany?
- How does air pollution in Germany behave around New Years?
- How did COVID-19 lockdowns and restrictions affect air pollution in Germany compared to the previous and following years?
- How does the air quality for PM_{10} vary between the different german federal states?
- How does air pollution for PM_{10} vary across urban, suburban and rural areas?
- How does air pollution vary between background, traffic and industrial monitoring stations?
- How do seasonal changes influence the pollution concentrations in Germany between 2016 and 2025?

Research Questions - other sources

- How did temperature and precipitation influence PM_{10} and $PM_{2.5}$ concentrations in major German cities between 2016 and 2025?
- How did summer school holidays in German federal states affect PM_{10} and $PM_{2.5}$ concentrations over the past ten years?
- How does Germany compare to other countries in terms of the daily and monthly average concentrations of PM_{10}, $PM_{2.5}$ and NO_2?

Data Collection

Data Collection

- Main data source: Umweltbundesamt (UBA) for pollution data
- Different monitoring stations placed across Germany
- Initially request data from all stations, then filter based on the question
  - Stations have information such as state, location, area type
  - Over 1.5 million requests for daily station data over a span of 10 years
- Additional data sources:
  - open-Meteo - weather data
  - openaq - international pollution data
  - public information for school holiday dates

Used Data Attributes

- Station identification: station id, station code, station name
- Location information: city, street, street number, ZIP code, longitude, latitude
- Operational information: active from, active to
- Network information: network id, network code, network name
- Station classification: station type, station setting (e.g. urban, suburban, rural)
- Weather data: peak temperature, precipitation sum

Challenges

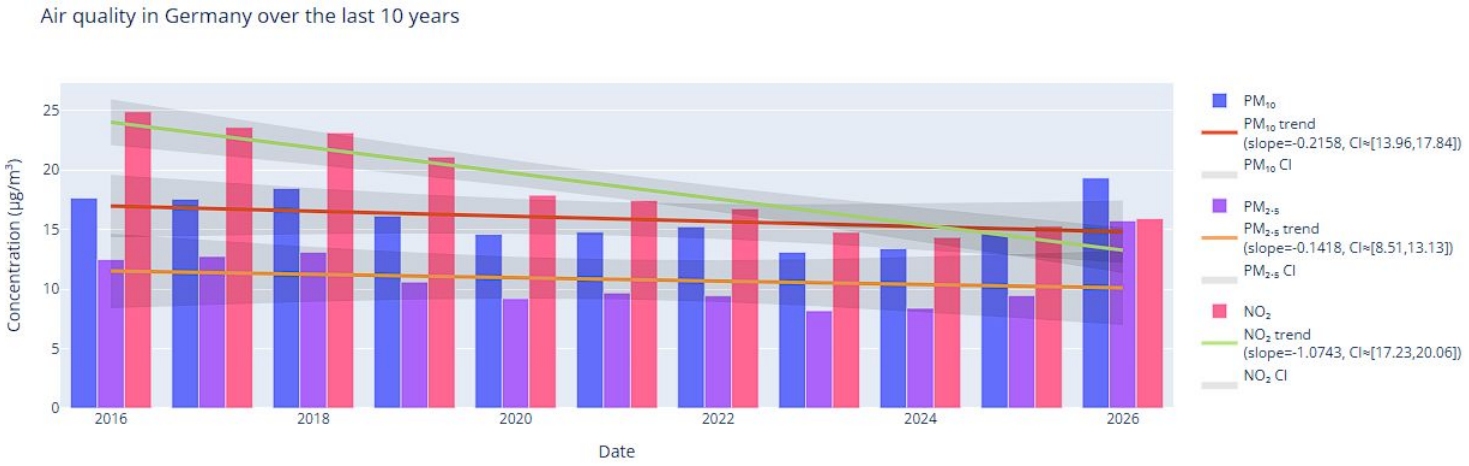
Challenges

- API rate limits when we needed large amounts of data
  - Request data in batches, add explicit wait timers
- Missing data from monitoring stations for certain pollutants
  - Case by case basis, sometimes mark missing data
- Styling the website with dash

Research Questions

Research Questions - Trends in Germany

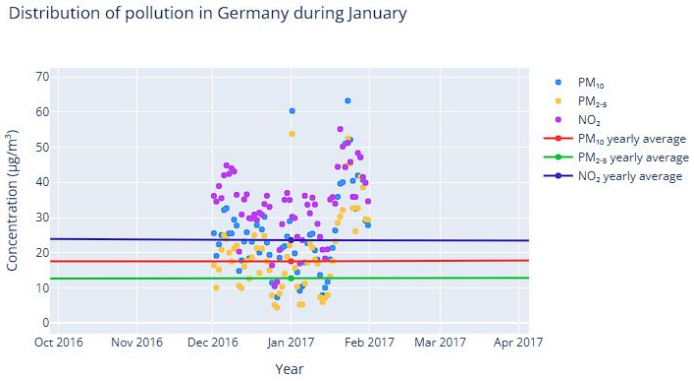
How did the average values of PM_{10}, $PM_{2.5}$ and NO_2 change in Germany?

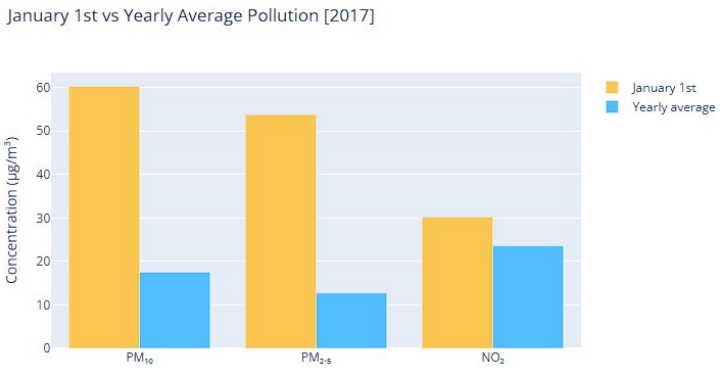


- A slight decrease in fine dust and a significant decrease in NO_2 suggest that air quality has improved over the past 10 years

Research Questions - New Years

How does air pollution in Germany behave around New Years?

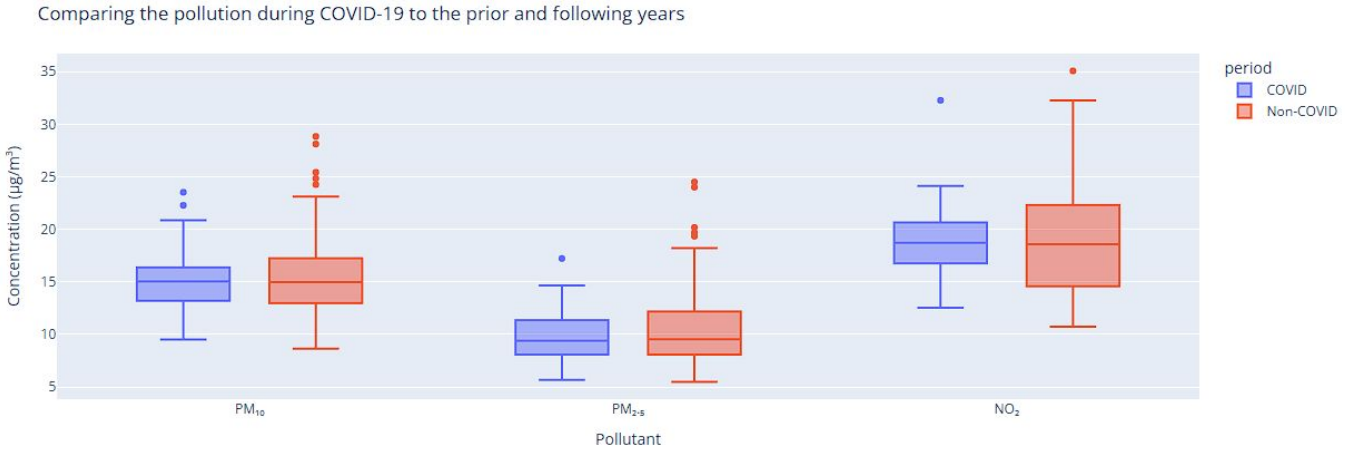




- Significant spikes in PM_{10} and $PM_{2.5}$ (especially on January 1 due to fireworks), while NO_2 was only slightly higher than the annual average

Research Questions - COVID-19

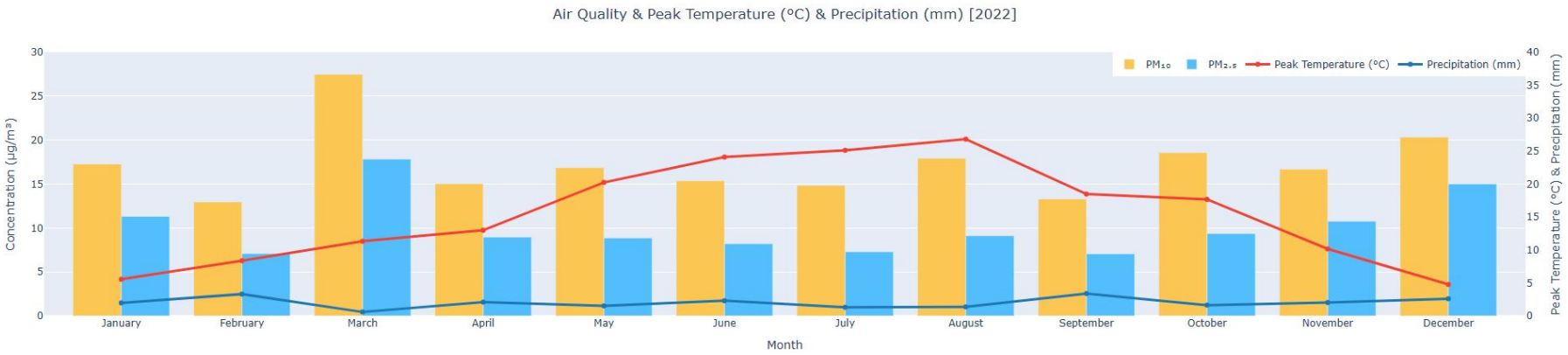
How did COVID-19 lockdowns and restrictions affect air pollution in Germany compared to the previous and following years?



- There are hardly any noticeable differences between the COVID and non-COVID periods, as the distributions and median values of the pollutants largely overlap

Research Questions - Weather

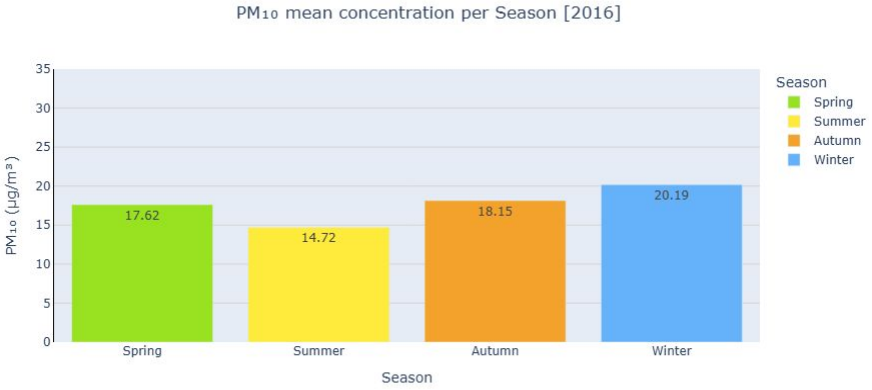
How did temperature and precipitation influence PM_{10} and $PM_{2.5}$ concentrations in major German cities between 2016 and 2025?

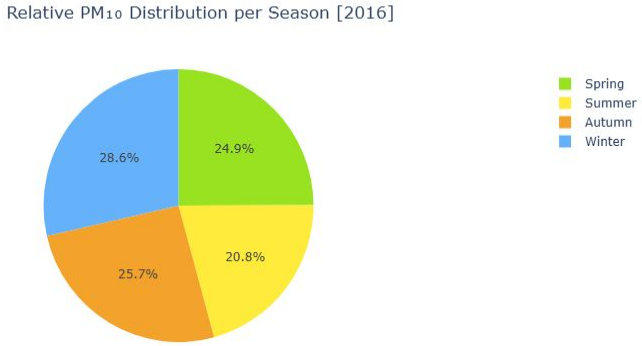


- Peak temperature and precipitation seem to have no impact on air pollution

Research Questions - Seasons

How do seasonal changes influence the pollution concentrations in Germany between 2016 and 2025?

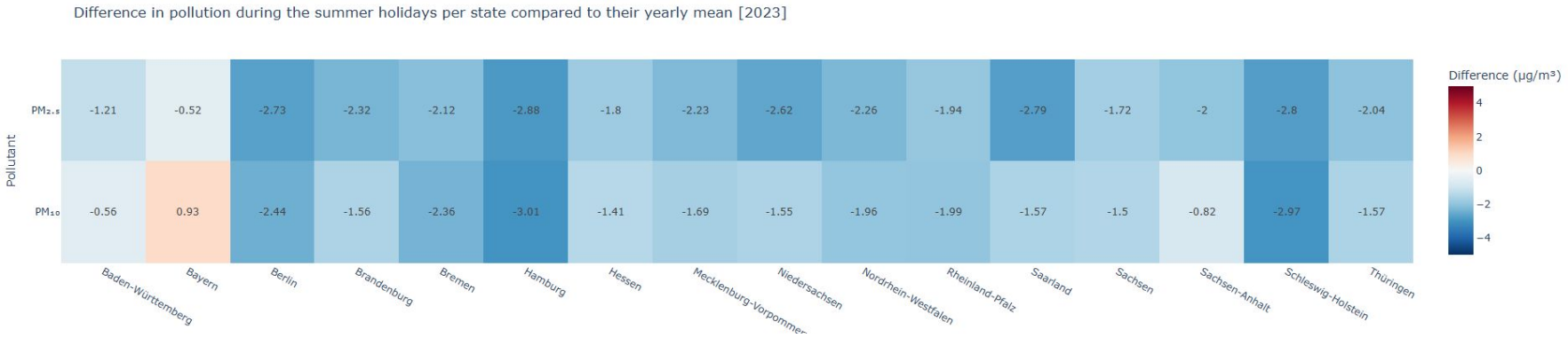




- Winter consistently more polluted, summer relatively clean

Research Questions - Holidays

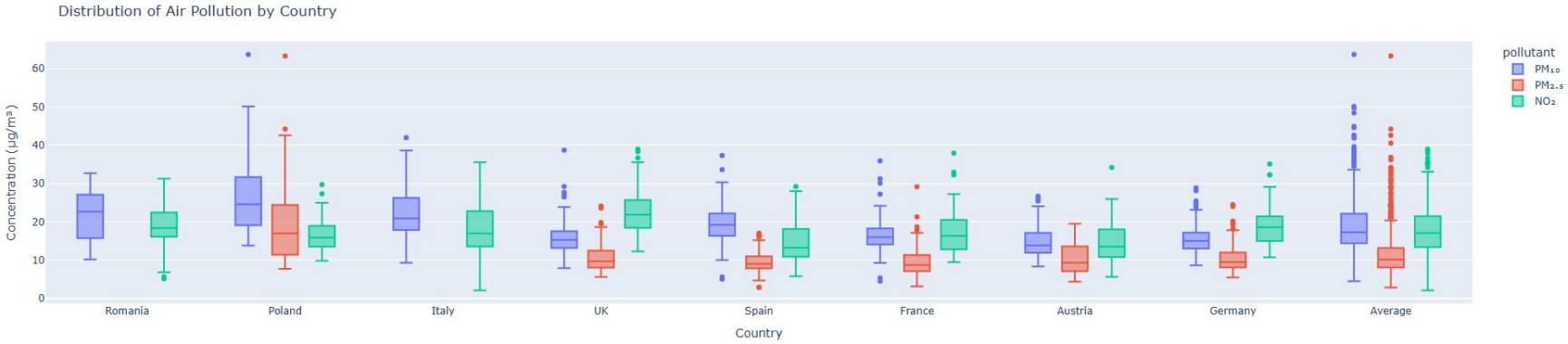
How did summer school holidays in German federal states affect PM_{10} and $PM_{2.5}$ concentrations over the past ten years?



- School holidays consistently cleaner, suggesting people leave the country or don't commute much / stay at home

Research Questions - Countries

How does Germany compare to other countries in terms of the average concentrations of PM_{10}, $PM_{2.5}$, and NO_2?



- In comparison, Germany's avg. PM_{10} and $PM_{2.5}$ values are below average, while the avg. NO_2 value is above average

Research Questions - States

How does the air quality vary between the different german federal states?

- Very similar pattern across all federal states for PM_{10}, $PM_{2.5}$, and NO_2

Website

Website

Live demo!

Thank you!